

# Workforce Attrition Intelligence Dashboard

## Employee Retention Risk & HR Analytics

Python · MySQL · Power BI Desktop

238.37bn Deposits · 1,470 Employees · 32 Columns · 16.12% Attrition · 3 Tools

| TOOLS                     | SOURCE                    | RECORDS         | DATE RANGE   | ACCOUNTS               |
|---------------------------|---------------------------|-----------------|--------------|------------------------|
| Python + MySQL + Power BI | IBM HR Analytics (Kaggle) | 1,470 · 32 Cols | Full Dataset | N/A — HR Employee Data |

### 01 EXECUTIVE SUMMARY

- Executed a 3-phase end-to-end analytics pipeline on IBM's 1,470-employee HR dataset — Phase 1 (Python): inspected 35 columns, dropped 3 zero-value constants, engineered 4 derived features (AgeGroup, IncomeBucket, TenureBucket, AttritionBin), and generated 10 EDA visualizations covering every major attrition risk dimension before exporting a clean 32-column CSV for MySQL.
- Phase 2 (MySQL): created a 32-column normalized table, imported data with inline Yes/No→1/0 conversion for Attrition and OverTime, built 8 performance indexes, executed 8 validated analytical queries establishing company-wide KPIs, and created vw\_hr\_summary — an analytical VIEW pre-aggregating 15 demographic and behavioral dimensions for Power BI multi-factor analysis.
- Phase 3 (Power BI): connected to MySQL, renamed tables in Power Query, engineered 12 DAX measures, and built a 3-page 26-visual interactive dashboard surfacing that Sales Representatives carry 39.76% attrition (2.47× company average), overtime workers leave at 2.93× the rate of non-overtime workers, the Single + Overtime + Job Level 1 combination reaches 67.80%, and employees who left earned \$2,046/month less than those who stayed — actionable retention intelligence across 6 business dimensions.

### 02 PROBLEM STATEMENT

IBM's HR department maintained records on 1,470 employees across 35 attributes — spanning compensation, tenure, job level, satisfaction scores, overtime status, stock options, and demographic data — with zero structured reporting layer. HR business partners had no visibility into which departments, job roles, or demographic groups were driving the company's 16.12% attrition rate. There was no mechanism to quantify the financial exposure from ongoing turnover, identify the highest-risk employee segments before they exited, or surface the behavioral and compensation patterns that distinguished the 237 employees who left from the 1,233 who stayed. Talent decisions were being made without data-backed insights while attrition continued silently across all 3 departments — with replacement cost exposure estimated at 50–200% of annual salary per exited employee, representing a potential \$13.6M–\$54.3M annual cost burden based on the \$4,787 average income of leavers.

### 03 OBJECTIVES

| #     | Objective                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Obj 1 | Execute a structured Python EDA pipeline — inspect, clean, engineer features, visualize 10 attrition risk dimensions, export clean CSV                |
| Obj 2 | Build a normalized MySQL table with inline Yes/No→1/0 conversion, 8 performance indexes, and 8 validated analytical queries                           |
| Obj 3 | Create vw_hr_summary VIEW aggregating 15 dimensions (AgeGroup, IncomeBucket, TenureBucket, HasOverTime, JobLevel, StockOptionLevel etc.)              |
| Obj 4 | Engineer 12 DAX measures enabling dynamic attrition rates, income gap quantification, and overtime impact calculation                                 |
| Obj 5 | Build a 3-page interactive Power BI dashboard with 26 visuals covering workforce overview, attrition drivers, and multi-factor risk segmentation      |
| Obj 6 | Surface actionable retention intelligence identifying highest-risk segments by role, tenure, overtime, income, marital status, and stock option level |

## 04 DATASET OVERVIEW

### Source & Structure

| Attribute       | Detail                                                                   |
|-----------------|--------------------------------------------------------------------------|
| Dataset Name    | IBM HR Analytics Employee Attrition Dataset                              |
| Source          | Kaggle — pavansubhasht/ibm-hr-analytics-attrition-dataset                |
| File Format     | CSV (.csv)                                                               |
| Raw Records     | 1,470 rows · 35 columns                                                  |
| Clean Records   | 1,470 rows · 32 columns (3 zero-value constants dropped)                 |
| Null Values     | Zero — 100% complete, no imputation required                             |
| Duplicate Rows  | Zero — EmployeeNumber unique across all 1,470 records                    |
| Attrition Split | 237 Left (16.12%) · 1,233 Stayed (83.88%)                                |
| Departments     | 3 — Research & Development (961), Sales (446), Human Resources (63)      |
| Job Roles       | 9 unique roles                                                           |
| Income Range    | \$1,009 – \$19,999 monthly                                               |
| Dropped Columns | EmployeeCount (always=1) · Over18 (always=Y) · StandardHours (always=80) |

### Column Definitions (32 Columns After Cleaning)

| Column         | MySQL Type      | Description          |
|----------------|-----------------|----------------------|
| EmployeeNumber | INT PRIMARY KEY | Unique employee ID   |
| Age            | TINYINT         | Employee age (18–60) |

| Column                   | MySQL Type  | Description                                                  |
|--------------------------|-------------|--------------------------------------------------------------|
| Attrition                | TINYINT(1)  | 1=Left · 0=Stayed (converted from Yes/No)                    |
| BusinessTravel           | VARCHAR(20) | Non-Travel / Travel_Rarely / Travel_Frequently               |
| DailyRate                | INT         | Daily compensation rate                                      |
| Department               | VARCHAR(30) | Sales / Research & Development / Human Resources             |
| DistanceFromHome         | TINYINT     | Distance from home (1–29)                                    |
| Education                | TINYINT     | Education level (1=Below College → 5=Doctor)                 |
| EducationField           | VARCHAR(25) | Life Sciences / Medical / Marketing / Technical / HR / Other |
| EnvironmentSatisfaction  | TINYINT     | Environment rating (1=Low → 4=Very High)                     |
| Gender                   | VARCHAR(10) | Male / Female                                                |
| HourlyRate               | TINYINT     | Hourly pay rate                                              |
| JobInvolvement           | TINYINT     | Job involvement rating (1=Low → 4=Very High)                 |
| JobLevel                 | TINYINT     | Job level tier (1=Entry → 5=Executive)                       |
| JobRole                  | VARCHAR(35) | 9 unique roles                                               |
| JobSatisfaction          | TINYINT     | Job satisfaction rating (1=Low → 4=Very High)                |
| MaritalStatus            | VARCHAR(10) | Single / Married / Divorced                                  |
| MonthlyIncome            | INT         | Monthly salary in USD                                        |
| MonthlyRate              | INT         | Monthly rate                                                 |
| NumCompaniesWorked       | TINYINT     | Companies worked at previously                               |
| OverTime                 | TINYINT(1)  | 1=Overtime · 0=No Overtime (converted from Yes/No)           |
| PercentSalaryHike        | TINYINT     | Last salary hike percentage                                  |
| PerformanceRating        | TINYINT     | Performance (3=Excellent · 4=Outstanding only)               |
| RelationshipSatisfaction | TINYINT     | Relationship satisfaction (1=Low → 4=Very High)              |
| StockOptionLevel         | TINYINT     | Stock option tier (0=None → 3=High)                          |
| TotalWorkingYears        | TINYINT     | Total career experience in years                             |
| TrainingTimesLastYear    | TINYINT     | Training sessions attended last year                         |
| WorkLifeBalance          | TINYINT     | Work-life balance rating (1=Bad → 4=Best)                    |
| YearsAtCompany           | TINYINT     | Tenure at IBM in years                                       |
| YearsInCurrentRole       | TINYINT     | Years in current role                                        |
| YearsSinceLastPromotion  | TINYINT     | Years since last promotion                                   |
| YearsWithCurrManager     | TINYINT     | Years with current manager                                   |

## 05 TOOLS & TECHNOLOGIES

| Tool / Technology     | Purpose in This Project                                                              |
|-----------------------|--------------------------------------------------------------------------------------|
| Python 3.11           | Phase 1 — EDA, 3 column drops, 4 feature creations, 10 chart generations, CSV export |
| pandas                | pd.read_csv(), df.drop(), pd.cut() binning, groupby(), value_counts(), mean()        |
| numpy                 | Array operations, np.arange() for bar chart positioning                              |
| matplotlib + seaborn  | 10 EDA visualizations — donut, bar, horizontal bar, box plot, grouped bar            |
| MySQL 8.0 + Workbench | Phase 2 — Table creation, LOAD DATA LOCAL INFILE, 8 indexes, 8 queries, VIEW         |
| Power BI Desktop      | Phase 3 — MySQL connection, Power Query cleanup, DAX measures, 3-page dashboard      |
| Power Query Editor    | Table renaming (removed database prefix), data type verification                     |
| DAX                   | 12 calculated measures for KPIs, attrition rates, income gap, overtime analysis      |
| JSON Theme File       | HR Intelligence — Workforce Analytics Pro (custom deep navy + teal theme)            |

## 06 PYTHON EDA PROCESS (10 Steps)

| Step | Action                                                                          | Result / Impact                                            |
|------|---------------------------------------------------------------------------------|------------------------------------------------------------|
| 01   | pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv', encoding='utf-8-sig')      | Shape (1470, 35) confirmed                                 |
| 02   | df.isnull().sum() + df.duplicated().sum()                                       | Zero nulls · Zero duplicates                               |
| 03   | df.drop(columns=['EmployeeCount', 'Over18', 'StandardHours'])                   | Removed 3 zero-value constants → 32 columns                |
| 04   | pd.cut(df['Age'], bins=[18,25,30,35,40,45,50,60], labels=[...])                 | AgeGroup: 7 buckets created                                |
| 05   | pd.cut(df['MonthlyIncome'], bins=[0,3000,6000,10000,15000,21000], labels=[...]) | IncomeBucket: 5 ranges created                             |
| 06   | pd.cut(df['YearsAtCompany'], bins=[0,1,2,5,10,15,20,40], labels=[...])          | TenureBucket: 7 ranges created                             |
| 07   | (df['Attrition'] == 'Yes').astype(int)                                          | AttritionBin: Boolean→0/1 for correlations                 |
| 08   | Generated 10 EDA charts using matplotlib + seaborn — saved as PNG               | 10 visualization files saved                               |
| 09   | Dropped Python-only derived columns before export                               | AgeGroup, IncomeBucket, TenureBucket, AttritionBin removed |
| 10   | df_export.to_csv('hr_attrition_cleaned.csv', index=False)                       | 1,470 rows · 32 columns · 0 nulls exported                 |

## 10 EDA Charts Generated

| Chart    | Type           | Key Finding Visualized                                                    |
|----------|----------------|---------------------------------------------------------------------------|
| Chart 1  | Donut          | 83.88% Stayed vs 16.12% Left overall split                                |
| Chart 2  | Horizontal Bar | Sales Rep 39.76% (red) → Research Director 2.50% (green)                  |
| Chart 3  | Grouped Bar    | OT workers 30.53% vs Non-OT 10.44% — side by side                         |
| Chart 4  | Box Plot       | Median income gap: Left=\$3,202 vs Stayed=\$5,204                         |
| Chart 5  | Bar            | Age 18-24 = 39.18% (red spike) — youngest age group highest risk          |
| Chart 6  | Bar            | Single=25.53% (red) vs Married=12.48% (orange) vs Divorced=10.09% (green) |
| Chart 7  | Horizontal Bar | Leavers score lower on ALL 5 satisfaction metrics                         |
| Chart 8  | Bar            | First 2 years = 34-36% danger zone (red), drops after year 5              |
| Chart 9  | Bar            | Frequent travelers 24.91% vs Non-Travel 8.00% — 3.11× gap                 |
| Chart 10 | Bar            | No stock options 24.41% vs Level 1 only 9.40% — 2.60× gap                 |

## 07 MYSQL PROCESS (13 Steps)

| Step | Action                                                                                                         | Result / Impact                                                              |
|------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 01   | CREATE DATABASE hr_project; CREATE TABLE hr_attrition (...)                                                    | 32-column table · EmployeeNumber as PRIMARY KEY                              |
| 02   | LOAD DATA LOCAL INFILE with SET clause for Attrition and OverTime                                              | Yes/No → 1/0 converted at import time                                        |
| 03   | SELECT COUNT(*) FROM hr_attrition                                                                              | 1,470 rows confirmed                                                         |
| 04   | SELECT Attrition, OverTime LIMIT 5                                                                             | 0/1 values confirmed                                                         |
| 05   | CREATE INDEX<br>idx_attrition/idx_dept/idx_jobrole/idx_overtime/idx_marital/idx_joblevel/idx_income/idx_tenure | 8 indexes for query performance                                              |
| 06   | Query 1 — Overall KPIs                                                                                         | 1470 employees · 237 left · 16.12% · \$6,503 avg · 7.0yr · 2.73 satisfaction |
| 07   | Query 2 — Attrition by Department                                                                              | Sales=20.63% · HR=19.05% · R&D=13.84%                                        |
| 08   | Query 3 — Attrition by Job Role                                                                                | Sales Rep=39.76% (highest) → Research Director=2.50% (lowest)                |
| 09   | Query 4 — Overtime Impact                                                                                      | Works OT=30.53% · No OT=10.44% (2.93× multiplier)                            |
| 10   | Query 5 — Income Bucket vs Attrition                                                                           | Under \$3K=28.61% → Over \$15K=3.76% (clear inverse)                         |
| 11   | Query 6 — Age Group Analysis                                                                                   | 18-24=39.18% (highest) · 45-49=9.22% (lowest mid-career)                     |

| Step | Action                                     | Result / Impact                                                                     |
|------|--------------------------------------------|-------------------------------------------------------------------------------------|
| 12   | Query 7 — Multi-Factor Risk Combinations   | Single+OT+Level1=67.80% (highest risk combo)                                        |
| 13   | Query 8 — CREATE VIEW vw_hr_summary AS ... | 15-dimension VIEW with attrition_rate_pct, avg_income, avg_satisfaction per segment |

## 08 POWER BI PROCESS (13 Steps)

| Step | Action                                                                        | Result / Impact                                             |
|------|-------------------------------------------------------------------------------|-------------------------------------------------------------|
| 01   | Get Data → MySQL Database → localhost · hr_project                            | Live MySQL connection established                           |
| 02   | Navigator → selected hr_attrition + vw_hr_summary → Transform Data            | Both tables loaded into Power Query                         |
| 03   | Renamed hr_project hr_attrition → hr_attrition                                | Database prefix removed — DAX measures resolve correctly    |
| 04   | Renamed hr_project vw_hr_summary → vw_hr_summary                              | Prefix removed — VIEW fields accessible cleanly             |
| 05   | Verified Attrition and OverTime as Whole Number (0/1) · Text columns verified | Data types confirmed correct                                |
| 06   | Close & Apply                                                                 | 1,470 rows loaded · VIEW loaded                             |
| 07   | Created 12 DAX measures under hr_attrition table                              | All measures resolve without error                          |
| 08   | Built Page 1 — Workforce Overview                                             | Title + 6 KPI cards + donut + 2 bar charts + 2 slicers      |
| 09   | Built Page 2 — Attrition Deep Dive                                            | Title + 6 analysis charts + 2 slicers                       |
| 10   | Built Page 3 — Risk Factor Analysis                                           | Title + KPI card + risk matrix table + 3 charts + 2 slicers |
| 11   | View → Sync Slicers → synced Department + Gender across Pages 1, 2, 3         | Cross-page filtering enabled                                |
| 12   | Applied HR Intelligence — Workforce Analytics Pro JSON theme                  | Professional deep navy + teal dashboard aesthetic           |
| 13   | Ctrl+S → HR_Attrition_Intelligence_Dashboard.pbix                             | Dashboard saved                                             |

## 09 DAX MEASURES (12 Measures)

| Measure Name    | DAX Formula                  | Result |
|-----------------|------------------------------|--------|
| Total Employees | COUNTROWS(hr_attrition)      | 1,470  |
| Total Left      | SUM(hr_attrition[Attrition]) | 237    |

| Measure Name            | DAX Formula                                                  | Result      |
|-------------------------|--------------------------------------------------------------|-------------|
| Total Stayed            | [Total Employees] - [Total Left]                             | 1,233       |
| Attrition Rate %        | DIVIDE([Total Left], [Total Employees]) * 100                | 16.12%      |
| Avg Monthly Income      | AVERAGE(hr_attrition[MonthlyIncome])                         | \$6,503     |
| Avg Age                 | AVERAGE(hr_attrition[Age])                                   | 36.9        |
| Avg Tenure              | AVERAGE(hr_attrition[YearsAtCompany])                        | 7.0 years   |
| Avg Job Satisfaction    | AVERAGE(hr_attrition[JobSatisfaction])                       | 2.73 / 4.00 |
| Overtime Employees      | CALCULATE([Total Employees], hr_attrition[OverTime] = 1)     | 416         |
| Overtime Attrition Rate | CALCULATE([Attrition Rate %], hr_attrition[OverTime] = 1)    | 30.53%      |
| Income Left             | CALCULATE([Avg Monthly Income], hr_attrition[Attrition] = 1) | \$4,787     |
| Income Stayed           | CALCULATE([Avg Monthly Income], hr_attrition[Attrition] = 0) | \$6,833     |

Note: OverTime and Attrition stored as TINYINT(1) in MySQL. All CALCULATE() filters use = 1 / = 0 — not = "Yes" / = "No".

## 10 DASHBOARD VISUALS (26 Visuals Across 3 Pages)

### Page 1 — Workforce Overview

| #  | Visual        | Fields                        | Insight                                                    |
|----|---------------|-------------------------------|------------------------------------------------------------|
| 1  | KPI Card      | Total Employees               | 1,470 total workforce headcount                            |
| 2  | KPI Card      | Total Left                    | 237 employees exited the company                           |
| 3  | KPI Card      | Attrition Rate %              | 16.12% company-wide attrition rate                         |
| 4  | KPI Card      | Avg Monthly Income            | \$6,503 average monthly salary                             |
| 5  | KPI Card      | Avg Tenure                    | 7.0 years average company tenure                           |
| 6  | KPI Card      | Avg Job Satisfaction          | 2.73 / 4.00 overall satisfaction score                     |
| 7  | Donut Chart   | Attrition / Total Employees   | 83.88% Stayed (dark) vs 16.12% Left (light)                |
| 8  | Clustered Bar | JobRole / Total Employees     | Sales Executive #1 at 326 · Research Director lowest at 80 |
| 9  | Clustered Bar | Department / Attrition Rate % | Sales 20.63% · HR 19.05% · R&D 13.84%                      |
| 10 | Slicer        | Department (Dropdown)         | Human Resources / Research & Development / Sales           |
| 11 | Slicer        | Gender (Tile)                 | Female / Male                                              |

## Page 2 — Attrition Deep Dive

| #  | Visual           | Fields                                   | Insight                                            |
|----|------------------|------------------------------------------|----------------------------------------------------|
| 12 | Clustered Bar    | JobRole / Attrition Rate %               | Sales Rep 39.76% → Research Director 2.50%         |
| 13 | Clustered Bar    | OverTime / Attrition Rate %              | 1 (OT) = 30.53% vs 0 (Non-OT) = 10.44%             |
| 14 | Clustered Column | MaritalStatus / Attrition Rate %         | Single 25.53% > Married 12.48% > Divorced 10.09%   |
| 15 | Clustered Column | BusinessTravel / Attrition Rate %        | Frequent 24.91% > Rarely 14.96% > Non-Travel 8.00% |
| 16 | Clustered Bar    | AgeGroup / attrition_rate_pct (VIEW)     | 18-24 highest risk · 45-49 lowest                  |
| 17 | Clustered Bar    | IncomeBucket / attrition_rate_pct (VIEW) | Under \$3K highest · Over \$15K lowest             |
| 18 | Slicer           | OverTime (Range Slider 0–1)              | Toggle overtime vs non-overtime cohort             |
| 19 | Slicer           | MaritalStatus (Tile)                     | Divorced / Married / Single                        |

## Page 3 — Risk Factor Analysis

| #  | Visual           | Fields                                                                                           | Insight                                              |
|----|------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 20 | KPI Card         | Income Left                                                                                      | \$4,787 average income of employees who left         |
| 21 | Table            | Dept · JobRole · HasOverTime · MaritalStatus · attrition_rate_pct · total_employees · avg_income | Multi-factor risk matrix — highest risk combinations |
| 22 | Clustered Column | StockOptionLevel / Attrition Rate %                                                              | Level 0 = 24.41% · Level 1 = 9.40% · Level 3 = 18.0% |
| 23 | Clustered Bar    | TenureBucket / attrition_rate_pct (VIEW)                                                         | 2-4yr and 1yr = highest · 15-19yr = lowest           |
| 24 | Clustered Bar    | JobRole / Avg Job Satisfaction                                                                   | Healthcare Rep highest · Manager lowest              |
| 25 | Slicer           | Department (Dropdown)                                                                            | Filter entire page by department                     |
| 26 | Slicer           | JobLevel (Range Slider 1–5)                                                                      | Filter by job level tier                             |

# 11 KEY INSIGHTS & RESULTS

## A. Overall Attrition Profile

- Analyzed 1,470 IBM employees across 9 job roles and 3 departments — identifying 237 who left (16.12%) vs 1,233 who stayed (83.88%), with \$6,503 average monthly income, 7.0 years average tenure, and 2.73/4.00 average job satisfaction across the full workforce.



- Sales department leads attrition at 20.63%, followed by Human Resources at 19.05% and Research & Development at 13.84% — a 6.79 percentage-point spread confirming unequal attrition distribution across departments.

## B. Job Role Risk Segmentation

- Sales Representative ranked highest attrition role at 39.76% — 2.47× the company average — on \$2,626 average income and 2.9 years average tenure, signalling a structural frontline sales retention failure that independently accounts for 33 of 237 total exits.
- Research Director holds the lowest attrition at 2.50% on \$16,034 average income and 10.9 years tenure — confirming that seniority, income, and tenure duration are the strongest combined protection factors against departure.
- Laboratory Technician (23.94%), Human Resources (23.08%), and Sales Executive (17.48%) complete the top 4 risk roles — collectively accounting for 152 of 237 exits (64.1% of all attrition) while representing only 50.3% of total headcount.

## C. Overtime & Work-Life Impact

- Overtime workers leave at 30.53% vs non-overtime at 10.44% — a 2.93× attrition multiplier representing the single most powerful controllable driver in the dataset, affecting 416 of 1,470 employees (28.30% of the workforce).
- The Single + Overtime + Job Level 1 combination reaches 67.80% attrition — the highest multi-factor risk combination — with 59 employees in this segment of whom 40 have already exited, identifying a hyper-specific intervention target for HR retention strategy.
- Work-life balance scores confirm the compounding effect: leavers average 2.66 vs stayers at 2.78 — a consistent directional gap reinforcing that overtime's impact flows through perceived quality-of-life degradation.

## D. Compensation & Income Analysis

- Employees who left earned \$4,787 average monthly income vs \$6,833 for those who stayed — a \$2,046 monthly income gap (42.8% differential) that is the clearest single financial signal of attrition risk across the dataset.
- Under \$3,000/month employees face 28.61% attrition — nearly double the company average — while over \$15,000/month employees face only 3.76%, confirming a clear inverse income-attrition relationship consistent across all income buckets without exception.
- Median income gap: \$3,202 (left) vs \$5,204 (stayed) — a 62.5% difference — reinforcing that compensation is the primary lever for attrition reduction, particularly for frontline roles with the highest attrition rates.

## E. Demographic & Tenure Patterns

- Age 18–24 employees face 39.18% attrition — the highest of any age group — dropping sharply to 23.14% at 25–29, then 18.15% at 30–34, and stabilizing below 14% from age 35 onward. Early-career retention is the company's most urgent and largest demographic challenge.
- Year 1 attrition: 36.36% · Year 2 attrition: 34.50% — the first two years are the critical danger zone. Attrition falls to 11.07% for employees who survive to 5–9 years, confirming that if an employee crosses the 5-year threshold, retention probability increases dramatically.

- Single employees leave at 25.53% — more than double married (12.48%) and more than double divorced (10.09%) — a statistically significant marital status signal that likely correlates with age (younger = more single = more mobile) as a confounding factor.

## F. Stock Options, Travel & Satisfaction

- No stock options (Level 0) = 24.41% attrition vs Stock Level 1 = 9.40% — a 2.60× attrition reduction for employees with even a minimal equity stake. Level 3 (highest stock) returns to 18.0% — likely a tenure effect where senior employees have maximized vesting and are more mobile again.
- Frequent business travelers: 24.91% attrition vs non-travelers at 8.00% — a 3.11× gap identifying travel burden as a compounding retention risk factor, particularly when combined with overtime.
- Every satisfaction metric is lower for employees who left: Job Satisfaction: 2.47 vs 2.78 · Environment: 2.46 vs 2.77 · Relationship: 2.50 vs 2.71 · Work-Life Balance: 2.66 vs 2.78 · Job Involvement: 2.45 vs 2.77 — a consistent pattern confirming that attrition reflects systemic dissatisfaction, not isolated incidents.

## 12 KPI SUMMARY TABLE

| KPI                           | Value                  | KPI                     | Value                        |
|-------------------------------|------------------------|-------------------------|------------------------------|
| Total Employees               | 1,470                  | Total Exited            | 237 (16.12%)                 |
| Total Stayed                  | 1,233 (83.88%)         | Avg Monthly Income      | \$6,503                      |
| Avg Age                       | 36.9 years             | Avg Tenure              | 7.0 years                    |
| Avg Job Satisfaction          | 2.73 / 4.00            | Overtime Workers        | 416 (28.30%)                 |
| Highest Risk Role             | Sales Rep — 39.76%     | Lowest Risk Role        | Research Director — 2.50%    |
| Highest Risk Dept             | Sales — 20.63%         | Lowest Risk Dept        | R&D — 13.84%                 |
| OT Attrition Rate             | 30.53%                 | Non-OT Attrition        | 10.44%                       |
| OT Multiplier                 | 2.93×                  | Single Attrition        | 25.53%                       |
| Danger Combo (OT+Single+Lvl1) | 67.80%                 | OT+Single (any level)   | 49.62%                       |
| Age 18-24 Attrition           | 39.18%                 | Age 45-49 Attrition     | 9.22%                        |
| Year 1 Attrition              | 36.36%                 | Year 5-9 Attrition      | 11.07%                       |
| Avg Income — Left             | \$4,787                | Avg Income — Stayed     | \$6,833                      |
| Income Gap                    | \$2,046/month (+42.8%) | Median Gap              | \$3,202 vs \$5,204 (+62.5%)  |
| Under \$3K Attrition          | 28.61%                 | Over \$15K Attrition    | 3.76%                        |
| No Stock Option Attrition     | 24.41%                 | Stock Level 1 Attrition | 9.40%                        |
| Frequent Travel Attrition     | 24.91%                 | Non-Travel Attrition    | 8.00%                        |
| Travel Multiplier             | 3.11×                  | Job Satisfaction Gap    | 2.47 (left) vs 2.78 (stayed) |

## 13 CHALLENGES FACED & HOW OVERCOME

### Challenge 1 — Python PATH Fatal Launcher Error

- Problem: Running pip install in CMD returned 'Fatal error in launcher: Unable to create process using c:\program files\python\python.exe' — Python was installed at C:\Users\kumar\AppData\Local\Programs\Python\Python311\ but the system PATH pointed to a non-existent path, breaking the pip launcher entirely. Additionally, running setx PATH to fix this issue triggered a WARNING: The data being saved is truncated to 1024 characters — a Windows hard limit on setx that silently corrupted the existing PATH variable.
- Solution: Bypassed the broken launcher by running python -m pip install pandas numpy matplotlib seaborn — calling pip as a Python module directly through the confirmed executable found via where python. The PATH was then repaired permanently using the GUI method: System Properties → Environment Variables → User Variables → Path → Edit → New — adding Python311 and Python311\Scripts as separate entries, which has no character limit and does not truncate existing PATH entries.

### Challenge 2 — Power BI Table Database Prefix

- Problem: After connecting to MySQL via Get Data → MySQL Database, the Power BI Navigator loaded both tables with database-prefixed names: hr\_project hr\_attrition and hr\_project vw\_hr\_summary. All 12 DAX measures referencing hr\_attrition[column] immediately displayed a red underline syntax error because the actual table name in the model was hr\_project hr\_attrition (with a space), which DAX cannot resolve without explicit single-quote wrapping.
- Solution: Opened Power Query Editor (Home → Transform Data), right-clicked each table in the Queries pane on the left, selected Rename, and removed the hr\_project prefix — saving as hr\_attrition and vw\_hr\_summary. After Close & Apply, all 12 DAX measures resolved correctly with zero errors. The alternative fix (wrapping every table reference in single quotes in DAX) was rejected as it would have required editing all 12 measures and made every formula significantly harder to read.

### Challenge 3 — Attrition & OverTime Stored as Yes/No Text

- Problem: The raw CSV stores both Attrition and OverTime as string values ("Yes"/"No"), making them unusable for numerical aggregations (SUM, AVG) needed for attrition rate calculations in MySQL queries and Power BI DAX measures. A post-import UPDATE statement would have worked but is slower and adds an extra step.
- Solution: Applied inline conversion during LOAD DATA LOCAL INFILE using the @variable approach — loading the raw text into temporary variables (@att, @ot) and applying SET Attrition = IF(@att = 'Yes', 1, 0) and SET OverTime = IF(@ot = 'Yes', 1, 0) in the same import command. This stored both columns as TINYINT(1) from the moment of import, enabling SUM(Attrition) for total exits, AVG(Attrition) \* 100 for attrition rate, and CALCULATE([measure], OverTime = 1) in DAX — all working correctly without any additional transformation steps.

### Challenge 4 — Three Zero-Value Constant Columns

- Problem: EmployeeCount, Over18, and StandardHours each had exactly one unique value across all 1,470 rows (1, "Y", and 80 respectively). Including them in MySQL creates storage and index overhead; including them in Power BI pollutes the field pane with useless columns that can accidentally be dragged into visuals, producing misleading constant-value charts.
- Solution: Detected in Python Step 2 via df.nunique() inspection, which returned 1 for all three columns. Eliminated with df.drop(columns=['EmployeeCount', 'Over18', 'StandardHours'],

inplace=True) before any analysis or export. Because they were dropped from the CSV before MySQL import, they were never loaded into the database — reducing the table from 35 to 32 meaningful, analytically useful columns.

## Challenge 5 — vw\_hr\_summary Aggregation Inflation in Power BI

- Problem: The AgeGroup, IncomeBucket, and TenureBucket bar charts on Pages 2 and 3 — which use attrition\_rate\_pct from vw\_hr\_summary — display values on a 4K–6K scale instead of readable percentages. This occurs because the VIEW pre-aggregates by 15 dimensions simultaneously, and Power BI's default aggregation (SUM) sums the attrition\_rate\_pct values across all dimension-combinations within each displayed group, producing inflated totals rather than the true rate per category.
- Solution (interim): The relative ranking of segments remains correct despite the inflated scale — the bar with the highest 'Sum of attrition\_rate\_pct' still correctly corresponds to the highest-risk segment, making the charts directionally accurate for comparison purposes. The correct fix for a future iteration is to either (1) create AVERAGEX DAX measures that divide total exits by total employees within each bucket rather than reading from the pre-aggregated VIEW, or (2) create separate, single-dimension views for each bucket type, eliminating the multi-dimension grouping inflation.

## 14 SKILLS LEARNED & TAKEAWAYS

- Multi-Phase Analytics Pipeline Architecture: Experienced the full end-to-end analyst workflow across 3 distinct phases where each phase's output directly feeds the next — Python's clean CSV feeds MySQL's LOAD DATA INFILE, MySQL's vw\_hr\_summary VIEW feeds Power BI's charts, and Python's EDA findings validate MySQL's query results. Errors cascade across phases, making careful verification at each phase boundary critical.
- Python pd.cut() Feature Engineering: Applied supervised binning to continuous variables (Age, MonthlyIncome, YearsAtCompany) into analytically meaningful labelled buckets using pd.cut() — understanding the distinction between Python-only derived columns (used for visualization) and MySQL-reproduced buckets (created via CASE WHEN for SQL queries and VIEW).
- MySQL LOAD DATA INFILE Inline Transformation: Mastered the @variable + SET clause pattern for converting text flags to integers at import time — avoiding a separate UPDATE pass while achieving the same result in a single, atomic import command. This pattern is directly applicable to any binary Yes/No or True/False column in any dataset.
- Power Query Table Renaming for DAX Compatibility: Learned that MySQL connector imports include the database name as a prefix in the table name, and that this prefix must be removed in Power Query before any DAX measures are created — otherwise every measure formula requires explicit single-quote wrapping that reduces readability.
- CALCULATE() with Integer Filter Predicates: Built CALCULATE() measures filtering on integer columns (OverTime = 1, Attrition = 1) rather than text columns ("Yes", "No") — understanding that DAX filter predicates are type-dependent and that integer-stored binary flags require numeric equality operators.
- Multi-Source Power BI Dashboard (Table + VIEW): Built a single dashboard drawing simultaneously from a raw data table (hr\_attrition) and a pre-aggregated analytical VIEW (vw\_hr\_summary), requiring understanding of which visuals use which source and how Power BI handles two independent MySQL sources without an explicit relationship between them.
- Business-First Insight Communication: Translated 1,470 rows of transactional HR records into six categories of executive-actionable insights — quantifying the financial cost of the income gap (\$2,046/month differential), identifying the most dangerous employee combination (67.80% attrition), and sequencing findings from macro (department level) to micro (multi-factor risk segments) in the dashboard page structure.

## 15 CONCLUSION

- Executed a 3-phase Python → MySQL → Power BI pipeline on IBM's 1,470-employee HR dataset — dropping 3 useless constant columns, engineering 4 derived features, generating 10 EDA charts, building a 32-column MySQL table with inline Yes/No→1/0 conversion, running 8 validated analytical queries, creating a 15-dimension analytical VIEW, and producing a 3-page 26-visual Power BI dashboard — zero Excel required across the entire pipeline.
- Surfaced Sales Representatives at 39.76% attrition (2.47× company average), identified overtime as the single most controllable driver (30.53% vs 10.44% — 2.93× multiplier), quantified a \$2,046/month income gap between leavers and stayers, and pinpointed the first two years of tenure as the critical danger zone (34–36% attrition) — four independent, actionable retention levers discoverable only through structured multi-dimensional analysis.
- Delivered 12 DAX measures enabling dynamic cohort filtering across all risk dimensions — allowing HR to instantly isolate, for example, only single female sales representatives who work overtime and earn under \$3,000/month, quantify their exact attrition exposure, and build targeted retention actions for that precise segment within the same dashboard.
- Applied the complete advanced analytics toolkit — Python EDA and feature engineering, MySQL analytical querying and VIEW design, and Power BI DAX modelling and interactive dashboard construction — demonstrating end-to-end data analyst capability from raw CSV ingestion through executive-ready people intelligence, delivered in a single cohesive project.